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Explanation-Aware Incentives for Trustworthy Multi-Agent AI with Blockchain Provenance: Evidence from Pharmaceutical Supplier Selection and Peer-to-Peer Energy

Abstract When several AI agents decide together, each with its own objective, an agent can steer the group by attaching a misleading explanation to its vote, and ordinary payoffs neither reward faithful explanations nor penalise false ones. We propose an explanation-aware incentive in which explanation fidelity, measured against SHAP attributions of the agent's own model, evidence support, measured against observed data, and an audited manipulation score enter each agent's utility; a reputation score weights votes; and hashes of decisions, explanations, evidence, rewards and reputation are recorded on a local blockchain. We tested the mechanism under a pre-registered protocol on two real datasets: USAID/PEPFAR pharmaceutical deliveries (supplier selection) and an energy community with peer-to-peer trading, with five learning agents per domain and ten paired seeds. Manipulation fell in both domains (*-69.7%* and *-44.4%*). Explanation fidelity and evidence support improved strongly in the pharmaceutical case (*+83.5%*, *+75.7%*) but only negligibly in energy, where the baseline was already near its ceiling. The mechanism did not make the system more robust to a growing share of malicious agents. The provenance layer detected all 1,000 tampered records with no false positive in 29,800 unchanged ones. With open LLM agents, manipulation also fell, but a factorial experiment showed that their lower stated confidence followed the wording of the reward rules rather than the payoff applied. Explanation-aware incentives reduce strategic manipulation; their effect on explanation quality depends on the domain.

Keywords Multi-agent systems, Explainable AI, Mechanism design, Explanation fidelity, Reputation, Blockchain provenance

1 Introduction

Decisions that used to be taken by one model are increasingly split among several specialised AI agents, each responsible for one criterion: cost, reliability, grid stability, price. Their outputs are combined by a vote or a market, and each agent attaches an explanation to its contribution so that humans can audit the result. This arrangement creates an incentive problem that single-model explainability does not face. An agent whose private objective favours one option can vote for it and attach an explanation that sounds convincing but does not reflect what its model actually computed, or that cites evidence the data do not support. Explanation methods are known to be manipulable (Slack et al. 2020; Dombrowski et al. 2019; Heo et al. 2019), and plausible but unfaithful rationalisations can be constructed on purpose (Aïvodji et al. 2019). In a multi-agent decision, nothing in the usual payoff, which rewards only the decision, makes a faithful explanation worth more to the agent than a misleading one.

Mechanism design offers a way to make truthful reporting individually rational, from peer prediction (Miller et al. 2005) to the Bayesian truth serum (Prelec 2004), and incentive mechanisms are widely used to steer self-interested learners (Zhan et al. 2021). Reputation systems give persistent weight to agents that behaved well (Jøsang et al. 2007; Kamvar et al. 2003), and blockchains can make records of AI decisions tamper-evident (Liang et al. 2017; Nassar et al. 2019). Recent work combines blockchain with incentives for multi-agent learning (Tian 2025) or LLM agent collaboration (Kumala et al. 2026), and rewards agents for faithful explanations of a model (Cheng et al. 2026). To our knowledge, however, no study puts explanation fidelity and

evidence support, measured against the agent's own model and the observed data, directly into the utility of strategic agents that jointly take a decision, and tests the result on real operational data in more than one domain.

This paper does so. Our contributions are:

1. An explanation-aware incentive for multi-agent decisions in which each agent's utility combines decision quality, explanation fidelity, evidence support, cooperation and an audited manipulation score; a reputation term turns past rewards into voting weight; and a provenance layer records hashes of decisions, explanations, evidence, rewards and reputation on a blockchain (Section 3).
2. A pre-registered evaluation on two real datasets, pharmaceutical supplier selection and peer-to-peer energy, with a baseline, an ablation without reputation, malicious-agent and collusion experiments, a tampering experiment, cross-domain comparison and sensitivity analysis (Sections 4–5).
3. Evidence on what the mechanism does and does not achieve: it reduces manipulation in both domains; it improves explanation quality strongly in one domain and negligibly in the other; it does not improve robustness to malicious agents; and, with LLM agents, the behavioural change it produces is explained by the wording of the stated rules rather than by the payoff (Section 6).

2 Related work

Explanation fidelity. SHAP attributes a prediction to features with Shapley values (Lundberg and Lee 2017), and TreeSHAP computes them exactly for tree ensembles (Lundberg et al. 2020). Whether an explanation reflects the model is its fidelity; evaluation frameworks and metrics are surveyed by Doshi-Velez and Kim (2017) and Zhou et al. (2021), and Yeh et al. (2019) formalise infidelity and sensitivity. Miller (2019) argues that explanations for people must also be selective and contrastive, which is why our agents report only their top features. We use SHAP of each agent's own model as the reference against which a reported explanation is scored.

Explanation manipulation. Post-hoc explainers can be fooled: a model can hide its reliance on a sensitive feature from LIME and SHAP (Slack et al. 2020), saliency maps can be changed arbitrarily without changing predictions (Dombrowski et al. 2019; Heo et al. 2019), and an interpretable surrogate can rationalise an unfair model (Aïvodji et al. 2019). These works study an adversary who manipulates the explainer; we study agents who misreport their own explanation, and ask whether incentives can make that unprofitable.

Incentives and reputation. Peer prediction (Miller et al. 2005) and the Bayesian truth serum (Prelec 2004) reward truthful reports without ground truth; incentive mechanisms for federated learning reward contributions of self-interested participants (Zhan et al. 2021). Reputation systems aggregate past behaviour into trust (Jøsang et al. 2007), and EigenTrust does so in peer-to-peer networks (Kamvar et al. 2003). Our mechanism has ground truth of a particular kind available, the agent's own model and the observed data, and rewards agreement with it.

Blockchain for provenance and trustworthy AI. ProvChain records data provenance on a blockchain (Liang et al. 2017); Nassar et al. (2019) argue for blockchain in explainable and trustworthy AI. Tian (2025) combine blockchain with incentive-compatible multi-agent reinforcement learning, Kumala et al. (2026) propose blockchain-based reputation and incentives for LLM multi-agent collaboration, and Cheng et al. (2026) reward agents for faithful explanations. None of these puts the fidelity and evidence support of each agent's explanation into the utility of agents that vote on a shared decision and evaluates it on real data in two domains.

Application domains. Supplier selection in pharmaceutical supply chains is a multi-criteria problem (Forghani et al. 2018). Peer-to-peer energy trading and community markets are reviewed by Tushar et al. (2020), Sousa et

al. (2019) and Tushar et al. (2018), the latter for game-theoretic approaches; the Brooklyn Microgrid is a field case (Mengelkamp et al. 2018). Our energy data come from the study of García-Muñoz et al. (2025).

3 Method

Figure 1 shows the pipeline and Figure 2 the mechanism. All parameters below were fixed in a protocol frozen before the experiments; later changes were logged as dated amendments.

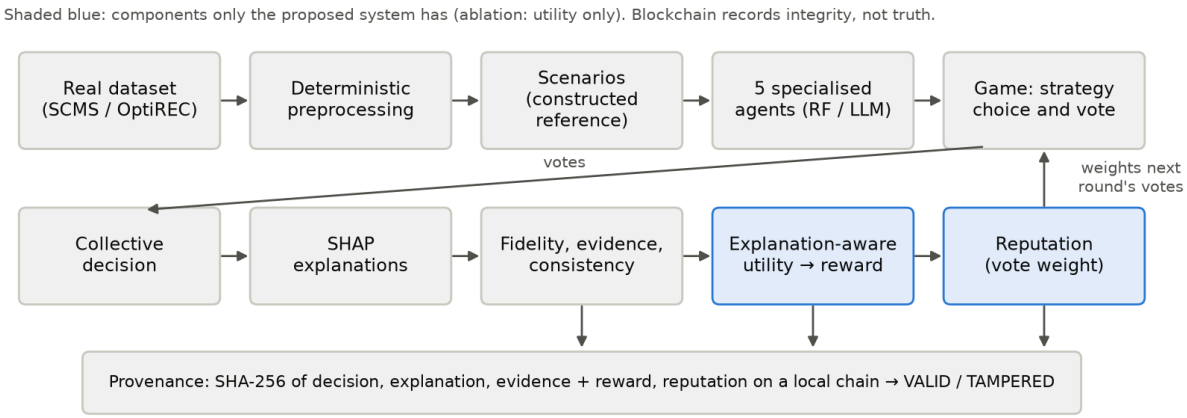


Fig. 1 Pipeline. Shaded blue: components that only the proposed system has (the ablation keeps the explanation-aware utility only). The provenance layer records integrity, not truth

3.1 Agents, actions and the game

Each domain has five agents $N=1,\ldots,5$ and a finite action set per scenario: three candidate manufacturing sites in the pharmaceutical case and \idle, charge, discharge\ for a community battery in the energy case. Agent i has a private objective $u_i(a)$, a closed-form function of real or derived variables (for example, the cost agent prefers the lowest expected unit price and the grid agent the smallest absolute grid exchange), normalised to $\tilde{u}_i(a)\in[0,1]$ over the actions of the scenario. Each agent also has a model, a random forest (Breiman 2001) with 100 trees of depth 6 trained on its own feature view to predict the reference decision (Section 4); the model's argmax is the agent's *model vote*.

In every round an agent submits a report: a vote, a confidence, an explanation of $k=2$ (feature, sign) pairs, and one evidence claim per explained feature stating its level (low, mid or high by training-set terciles). The strategy set has five elements. *Honest* reports the model vote, the model's probability, the top-k SHAP features with their true signs and true levels. *Strategic* switches to its privately preferred action when that gains at least 0.25 in $\tilde{u}_i$ and then inflates its confidence by 0.3, but explains truthfully. *Misattributing* and *overclaiming* vote like strategic but replace one explained feature by an unimportant one with a flipped sign, or one evidence level by a false one. *Manipulative* always votes for its preferred action with confidence 1 and reports the k least important features with flipped signs and false levels. These attacks change reports only; the observations are never altered.

The collective decision is a weighted confidence vote, $\text{argmax}_a \Sigma_i w_i c_i \mathbb{1}[v_i=a]$, with $w_i=1$ in the baseline and ablation and $w_i=Rep_i$ in the proposed system. Agents choose strategies with a stateless ε-greedy bandit ($\varepsilon=0.1$, learning rate 0.1) on the payoff

$$r_i = \tilde{u}_i(a^*) + \omega R_i, \qquad (1)$$

where a^* is the collective action and R_i the mechanism's reward. The first term is what agent i gains privately by steering the decision; ω sets how much that matters. We do not claim that the learning dynamics reach a Nash equilibrium; instead we report, for the all-honest profile, the payoff of every unilateral deviation.

3.2 Auditing explanations

The auditor has white-box access to each agent's model and computes exact TreeSHAP attributions φ for the voted action (Lundberg et al. 2020). Three per-round scores follow.

Fidelity is the share of the true top-k attribution mass that the agent reported with the correct sign,

$$F_i = \frac{\sum_{j \in \hat{e}_i,\ \mathrm{sign\ correct}} |\varphi_{ij}|}{\sum_{j \in \mathrm{top}\text{-}k(|\varphi_i|)} |\varphi_{ij}|},$$

with $F_i=0$ for an empty explanation and $F_i=1$ when all attributions are zero. *Evidence support* E_i is the share of the agent's claims whose stated level matches the observed value. *Manipulation* M_i is the mean of five audited components: false attribution $1\text{-}F_i$; confidence inflation $\max(0, c_i\text{-}p_i)$, where p_i is the model's probability of the vote; unsupported evidence $1\text{-}E_i$; strategic voting $\mathbb{1}[v_i \neq \text{model vote}]$; and collusion, flagged when an agent deviates from its model and another deviating agent casts the same vote. M_i is computed from reports and the audit only, never from knowledge of the strategy played. As secondary outcomes we report explanation consistency (Jaccard similarity of explanations across the five most similar scenarios), collective accuracy and manipulation success (the collective action is wrong and equals the vote of an agent that deviated from its model).

3.3 Explanation-aware utility, reward and reputation

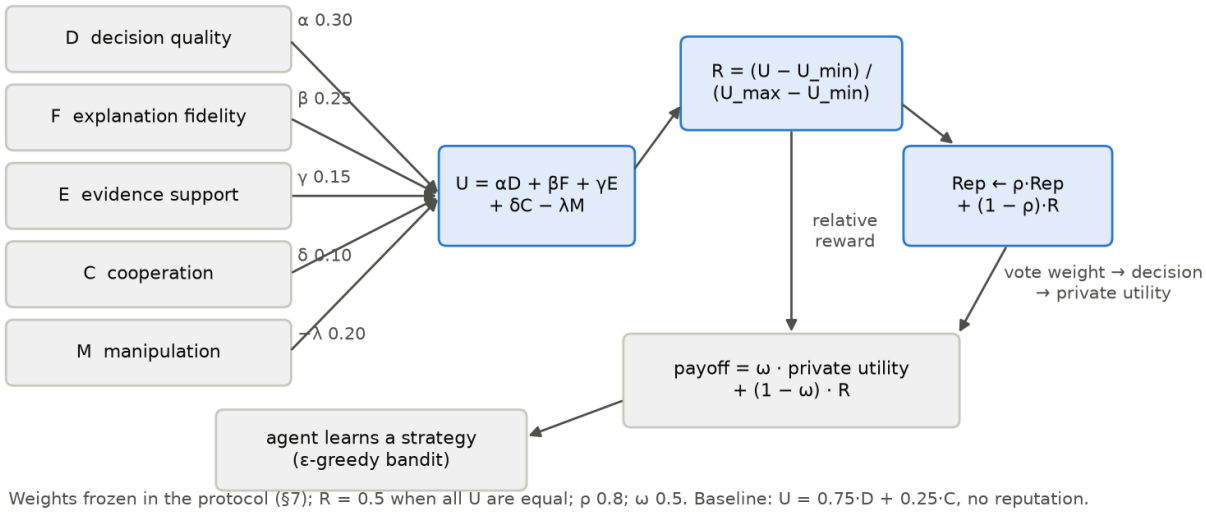


Fig. 2 The explanation-aware incentive. The utility combines five audited components; the reward is the utility rescaled across agents; reputation accumulates rewards and weights the next votes; the agent's payoff mixes its private utility of the collective action with the reward

The utility of agent i in a round is

$$U_i = \alpha D_i + \beta F_i + \gamma E_i + \delta C_i - \lambda M_i,$$

with $D_i=\mathbb{1}[v_i$ equals the reference decision$]$ and $C_i \in 0, 0.5, 1$ the agent's pivotal contribution (1 if removing it would turn a correct collective decision into a wrong one, 0 for the reverse, 0.5 otherwise). The weights $\alpha=0.30$, $\beta=0.25$, $\gamma=0.15$, $\delta=0.10$ and $\lambda=0.20$ were set a priori. The baseline uses $U_i=0.75D_i+0.25C_i$, the same ratio of decision to cooperation with no explanation terms. The reward rescales utilities across the agents of the round, $R_i=(U_i\text{-}U_{min})/(U_{max}\text{-}U_{min})$, with $R_i=0.5$ when all utilities are equal. Reputation starts at 0.5 and is updated as $Rep_i \leftarrow \rho\ Rep_i+(1\text{-}\rho)R_i$ with $\rho=0.8$, an effective memory of about five rounds.

We compare three systems on identical scenarios and seeds: the *baseline* (game and explanations, decision-only utility, equal votes), the *ablation* (explanation-aware utility, equal votes) and the *proposed* system (explanation-aware utility and reputation-weighted votes).

3.4 Provenance layer

For every agent and round, the provenance layer stores on chain the SHA-256 hashes of the canonically serialised decision, explanation and evidence, together with the reward and reputation scaled to integers ($\times 10^6$) and a block timestamp; the artefacts themselves stay off chain. A Solidity contract makes records append-only and writable only by the deploying account. Verification recomputes the hashes from the off-chain store and reports each record as valid, tampered or missing. The chain establishes that a record has not changed since it was written; it does not establish that the explanation was truthful.

This design trusts the operator who writes the records. A hardened variant, added after the main provenance results were known, removes part of that trust with three additions. First, *model registration*: each agent is bound once to a signing key and to the hash of its model identity (model, digest, decoding settings and the agent's objective). Second, *agent signatures*: the contract accepts a record only if the agent's registered key signed its identifier, artefact hashes and model hash for this chain and contract, so the operator cannot forge or replay an agent's output. Third, *anchoring*: Merkle roots (Laurie et al. 2013) of the registrations and of every 1,000 records are written to a second chain that the operator is assumed unable to rewrite, so a rebuilt history no longer matches the anchors. Rewards and reputations are computed by the mechanism and are therefore covered by the anchors but not by the agents' signatures.

3.5 LLM agents

As an extension, we replaced the five classical agents with an open LLM, Qwen2.5-7B (Qwen et al. 2025), served locally with Ollama at temperature 0. Each LLM agent receives the scenario, the output of its model (prediction and SHAP attributions) and the reward rules, and must write its own report in the fixed format above, which is audited exactly as before.

3.6 Use of generative AI

The experiment code, analysis scripts and a first draft of this manuscript were produced with an AI coding assistant (Anthropic's Claude, used through Claude Code) under the author's direction. All reported numbers are generated by that code from the saved results; tables and figures are produced by scripts from the result files; every reference was retrieved from Crossref, DataCite or arXiv by its identifier. The author reviewed the design, the protocol amendments and the text. Qwen2.5-7B served as the experimental subject in Section 3.5, not as a writing aid.

4 Experimental setup

Table 1 Case studies, scenarios and preprocessing

	Pharmaceutical	**Energy**
Source	USAID/PEPFAR SCMS delivery history (Development 2015)	OptiREC energy community (García-Muñoz 2025)
Scenario	one purchase order, three candidate sites	one hour of the community
Scenarios	1,648 (31 products, 49 sites)	8,760
Train / val / test	824 / 165 / 659 (by order date)	5,400 / 1,680 / 1,680 (week blocks)
Agents	cost, reliability, lead time, logistics, continuity	producer, consumer, market, grid, storage
Reference decision	equal-weight multi-criteria score (constructed)	daily perfect-foresight LP (constructed)
Simulated inputs	candidate set	battery: C-rate 0.5, efficiency 0.9
Derived variables	27	12

Pharmaceutical supplier selection. We use the USAID/PEPFAR Supply Chain Management System delivery history (Development 2015), 10,324 shipment line items of antiretroviral drugs and HIV test kits delivered

between 2006 and 2015 (Table 1). Only direct-from-supplier lines are used, so that delivery performance is the manufacturer's. A scenario is one purchase order for one product with three candidate manufacturing sites, the three that most recently delivered the product; the data contain no competing offers, so the candidate set is an experimental construction. Everything known about a site is computed from deliveries completed before the order date, and the split is chronological. The reference decision is the candidate with the highest equal-weight sum of five min–max normalised criteria (cost, delivery reliability, lead time, freight cost per kilogram, supply continuity), in the spirit of multi-criteria supplier selection (Forghani et al. 2018). It is a constructed reference: where the site actually used is a candidate, the reference picks it in 35% of scenarios, close to chance (33%). Agents correspond to the five criteria.

Peer-to-peer energy. We use one year of hourly operational data of a 55-user energy community under peer-to-peer trading and flexibility markets (García-Muñoz 2025; García-Muñoz et al. 2025). A scenario is one hour; the action concerns a community battery whose capacity is taken from the data and whose C-rate (0.5), round-trip efficiency (0.9) and state-of-charge window (10–90%) are simulated assumptions. The reference action is the solution of a daily perfect-foresight linear programme. Agents represent the producer, consumer, market, grid and storage perspectives. Splits use whole weeks in five-week cycles to avoid temporal leakage.

Runs and statistics. Each system is run with seeds 0–9 on identical scenarios (paired design): 600 rounds with evaluation on the last 300 in pharma, 1,000 and 500 in energy. The confirmatory family is fidelity (H1), evidence support (H2) and manipulation (H3), proposed versus baseline, tested per domain with a paired t-test or, when Shapiro–Wilk rejects normality, a Wilcoxon signed-rank test, with Holm correction (Holm 1979) across the three. The pre-registered decision rule calls an outcome *improved* if the Holm-adjusted $p<0.05$ in the hypothesised direction and the relative change $(proposed-baseline)/|baseline|$ is at least 5%, *worse* if significant in the opposite direction, and *negligible* otherwise. Robustness (H4) uses malicious shares of 0–40% (malicious agents always play manipulative) and compares the mean accuracy degradation over 10–40% between systems; collusion uses two colluding agents and three honest ones. Tamper detection (H5) is descriptive. Cross-domain generalisation (H6) holds for an outcome when both domains give the same verdict; domains are never pooled. The sensitivity analysis varies one parameter at a time: $\beta \in 0,0.1,0.25,0.4$, $\rho \in 0.5,0.8,0.95$ and $\omega \in 0.25,0.5,0.75$. LLM agents were run for 200 rounds with seeds 0–4.

5 Results

5.1 Explanation quality and manipulation

Table 2 Baseline vs proposed, classical agents ($n=10$ paired seeds; mean ± SD). Verdicts: pre-registered rule (Holm $p<0.05$ in the hypothesised direction and $|rel.| \geq 5\%$)

Domain	Metric	Baseline	Proposed	Δ [95% CI]	Rel.	p (Holm)	Verdict
pharma	Fidelity F	0.510 ± 0.077	0.936 ± 0.020	+0.426 [+0.370, +0.482]	+83.5%	*<0.0001*	improved
	Evidence support E	0.522 ± 0.070	0.917 ± 0.023	+0.395 [+0.345, +0.444]	+75.7%	*<0.0001*	improved
	Manipulation M	0.392 ± 0.038	0.119 ± 0.015	-0.273 [-0.302, -0.243]	-69.7%	*<0.0001*	improved
	Collective accuracy	0.578 ± 0.045	0.606 ± 0.033	+0.029 [-0.018, +0.076]	+5.0%	0.2021[a]	–
	Manipulation success	0.398 ± 0.057	0.263 ± 0.032	-0.135 [-0.185, -0.085]	-33.9%	0.0002[a]	–
energy	Fidelity F	0.936 ± 0.024	0.970 ± 0.004	+0.034 [+0.021, +0.050]	+3.6%	0.0039	negligible
	Evidence support E	0.926 ± 0.031	0.966 ± 0.004	+0.040 [+0.017, +0.063]	+4.3%	0.0039	negligible
	Manipulation M	0.066 ± 0.017	0.036 ± 0.005	-0.029 [-0.043, -0.016]	-44.4%	0.0026	improved
	Collective accuracy	0.744 ± 0.019	0.754 ± 0.013	+0.009 [-0.000, +0.019]	+1.2%	0.0541[a]	–
	Manipulation success	0.050 ± 0.017	0.025 ± 0.012	-0.025 [-0.037, -0.012]	-49.4%	0.0015[a]	–

[a]Secondary outcome, unadjusted p, no verdict.

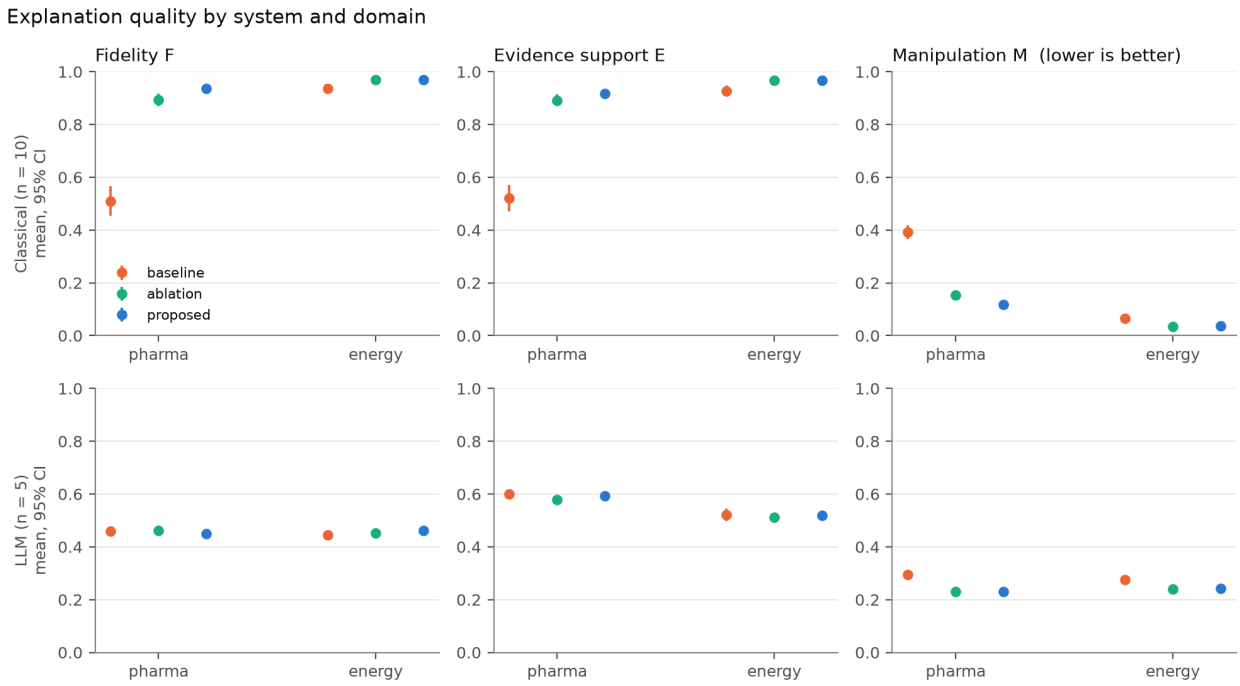


Fig. 3 Fidelity, evidence support and manipulation by system and domain, for classical (top) and LLM agents (bottom); points are means, bars 95% confidence intervals

Table 2 and Figure 3 show the main comparison. In the pharmaceutical case the incentive raised fidelity from 0.510 to 0.936 (*+83.5%*) and evidence support from 0.522 to 0.917 (*+75.7%*), and cut manipulation from 0.392 to 0.119 (*-69.7%*); all three are improved under the decision rule (Holm $p<0.0001$). In energy, manipulation fell from 0.066 to 0.036 (*-44.4%*, improved), while fidelity (*+3.6%*) and evidence support (*+4.3%*) rose significantly but below the 5% threshold, so both are negligible. The energy baseline already reached $F=0.936$ and $E=0.926$, leaving little room to improve; we report this as context and do not change the verdict. Collective accuracy did not change detectably in either domain, while manipulation success fell by a third in pharma and by half in energy (secondary outcomes). Explanation consistency rose by 28.6% and 8.6%.

The unilateral-deviation analysis qualifies these results. In energy, honesty is the best response for all five agents in every system, so the baseline is already close to honest. In the pharmaceutical case, honesty is the best response for no agent under the baseline and for two of five under the proposed system; the other three gain most from strategic voting with a truthful explanation. The incentive therefore removes the profit of false explanations without making the all-honest profile an equilibrium.

5.2 Ablation: incentive versus reputation

Table 3 Incentive ablation (incentive without reputation) vs baseline, and the descriptive increment of reputation (proposed - ablation)

Domain	Metric	Rel.	*p* (Holm)	Verdict	Reputation increment
pharma	Fidelity *F*	+74.9%	*<0.0001*	improved	+0.044
	Evidence support *E*	+70.9%	*<0.0001*	improved	+0.025
	Manipulation *M*	-61.0%	*<0.0001*	improved	-0.034
energy	Fidelity *F*	+3.5%	0.0037	negligible	+0.001
	Evidence support *E*	+4.4%	0.0037	negligible	-0.001
	Manipulation *M*	-46.2%	0.0013	improved	+0.001

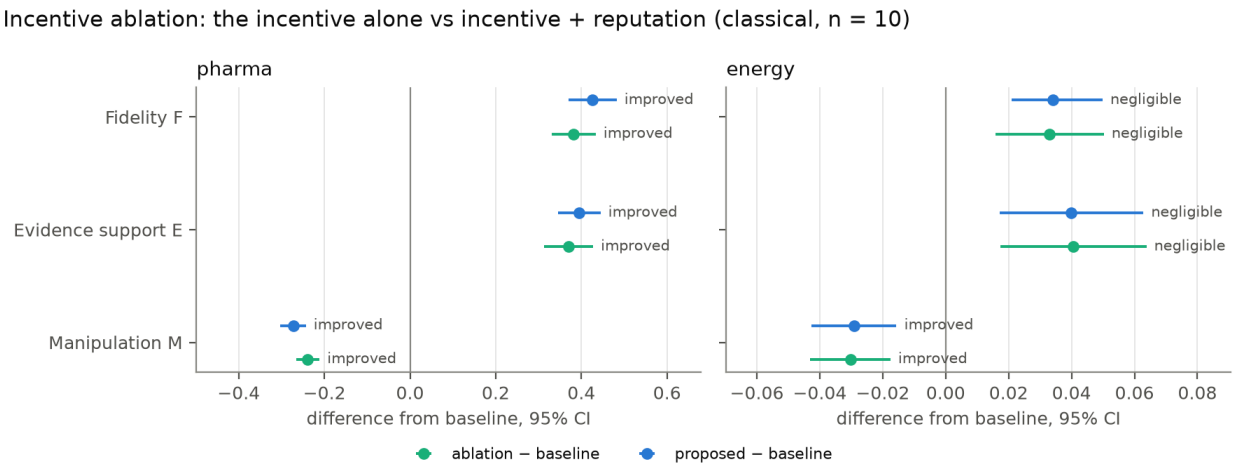


Fig. 4 Difference from the baseline for the ablation (incentive only) and the proposed system (incentive and reputation), with 95% confidence intervals and verdicts

The explanation-aware utility alone produces most of the effect (Table 3, Figure 4): the ablation reaches the same verdicts as the proposed system in both domains, with relative changes in pharma of *+74.9%* (fidelity), *+70.9%* (evidence) and *-61.0%* (manipulation). Reputation-weighted voting adds a small further gain in pharma (fidelity *+0.044*, manipulation *-0.034*) and none in energy; this comparison is descriptive.

5.3 Robustness to malicious agents and collusion

Table 4 Robustness. H4: mean accuracy degradation over 10–40% malicious agents, proposed - baseline (negative = proposed degrades less; paired *t*-test). Collusion: change in manipulation success (baseline / proposed) and in the colluders' reward under the proposed system when two agents collude (unadjusted)

Domain	H4 *Δ* [95% CI]	*p*	Verdict	Collusion: manip. success	Colluders' reward
pharma	+0.039 [-0.009, +0.087]	0.1021	not supported	+0.380 / +0.126	-0.458
energy	-0.020 [-0.043, +0.002]	0.0684	not supported	+0.155 / +0.047	-0.501

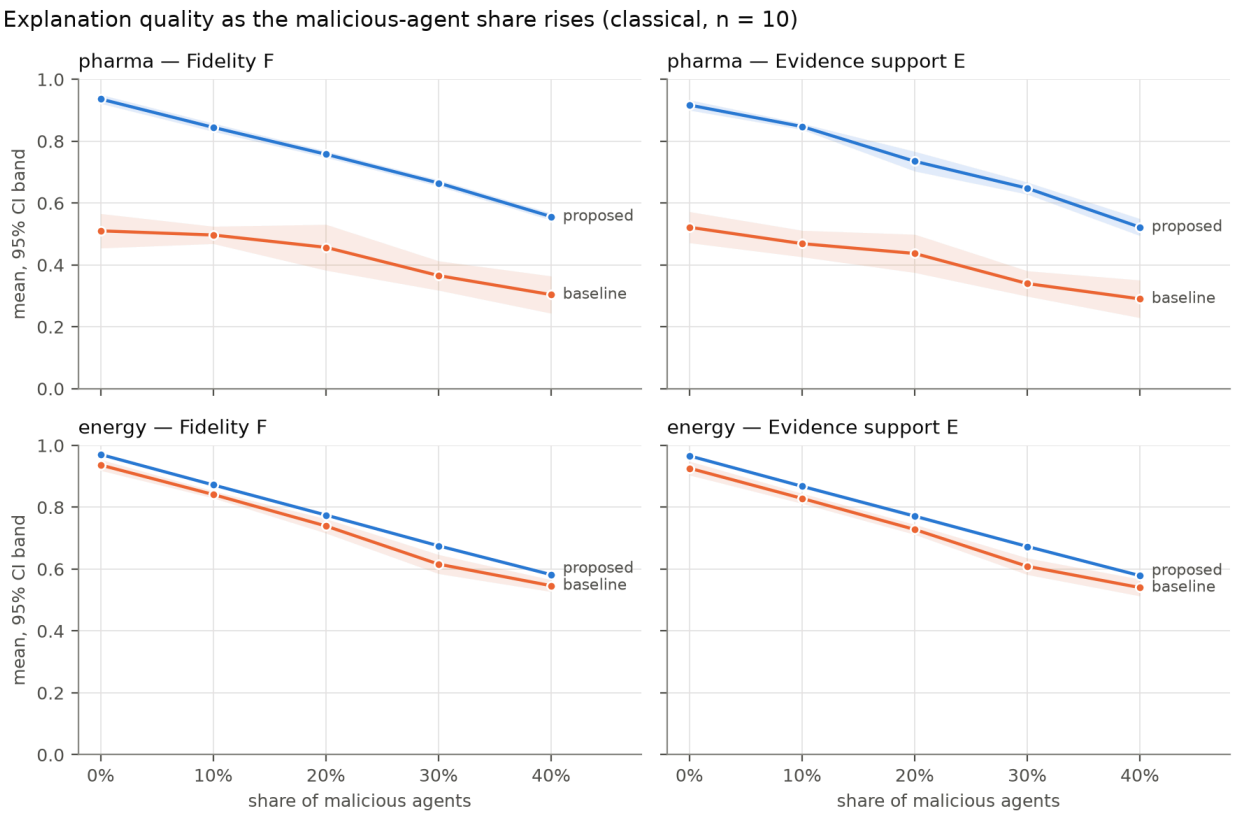


Fig. 5 Fidelity and evidence support as the share of malicious agents rises (means and 95% confidence bands)

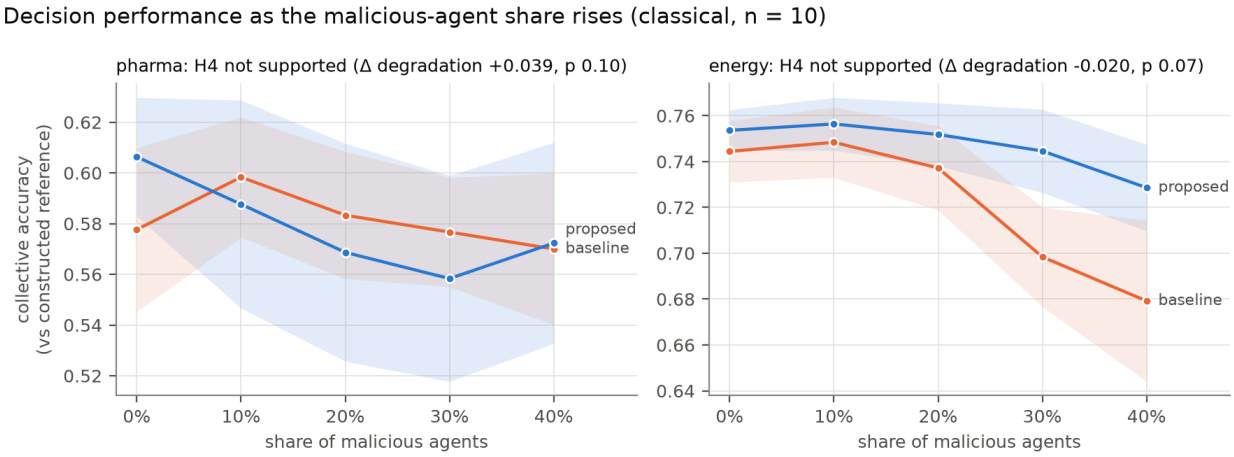


Fig. 6 Collective accuracy as the share of malicious agents rises

H4 is not supported (Table 4). The proposed system's accuracy did not degrade less than the baseline's as malicious agents rose from 10 to 40%: the mean difference in degradation was *+0.039* (*p=0.10*) in pharma and *-0.020* (*p=0.068*) in energy (Figure 6). The proposed system keeps clearly higher fidelity and evidence support than the baseline at every malicious share, but both fall roughly linearly as the share grows (Figure 5): the audit scores what malicious agents report, and the mechanism cannot make them report differently. Under collusion, the mechanism takes the reward away from the colluders (*-0.458* in pharma, *-0.501* in energy) and the honest agents' payoff rises; collusion still raises manipulation success, but by less than under the baseline (*+0.126* against *+0.380* in pharma; *+0.047* against *+0.155* in energy).

5.4 Tamper detection

Table 5 Tamper detection on the provenance layer (H5)

Group	Flagged TAMPERED	Meaning
decision	200 / 200	detected
explanation	200 / 200	detected
evidence	200 / 200	detected
reward	200 / 200	detected
reputation	200 / 200	detected
untouched controls	0 / 1,000	false positives
re-serialised controls	0 / 1,000	false positives
all other records	0 / 27,800	false positives
below 10^{-6} resolution	0 / 200	not detectable by design

Local node, 30,000 records. Batches of 100: 95,912 gas per record, 8.5 ms verification per record; 128 bytes on-chain per record.

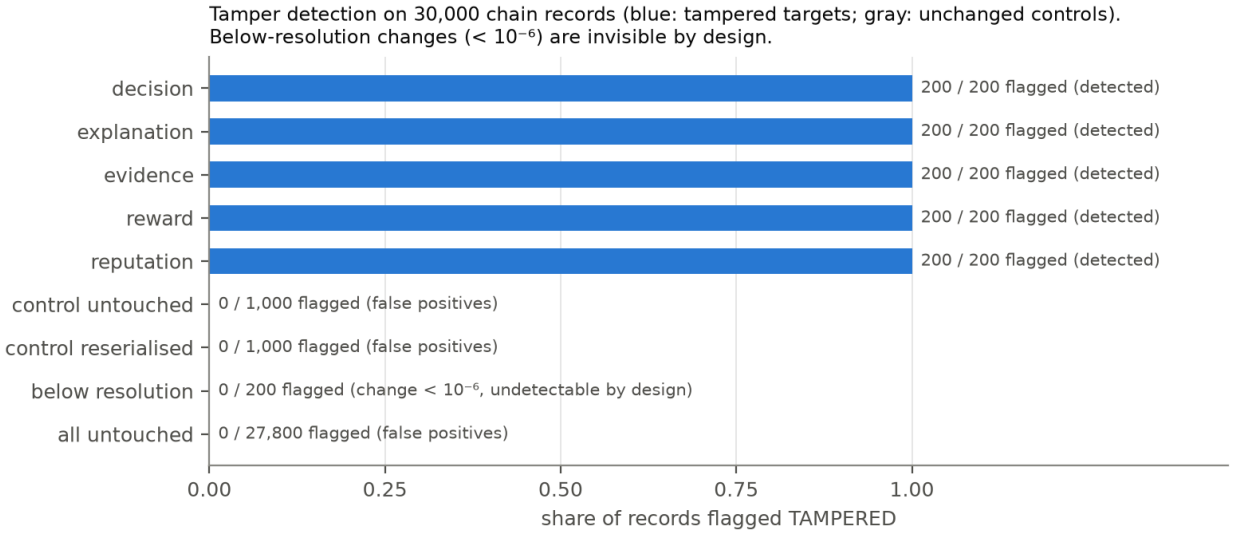


Fig. 7 Share of records flagged as tampered, by target and control group

The provenance layer recorded 30,000 agent-round records from the LLM main runs of both domains on a local Ethereum-compatible node. Of 1,000 records altered off chain, 200 in each of decision, explanation, evidence, reward and reputation, all were detected; no false positive occurred in 2,000 untouched or re-serialised controls or in the 27,800 other records (Table 5, Figure 7). Changes smaller than the 10^{-6} storage resolution of rewards and reputation are invisible by design. The contract refused all overwrites and writes from another account. In batches of 100, a record cost about 96,000 gas; local-node latency excludes consensus and network delay and will differ on a production network. H5 is supported in the sense of integrity: the layer shows that records were not changed, not that they were true.

Table 6 Hardened provenance: agent signatures, model registration and anchoring (amendment 9, descriptive)

Check	Result
Forged or replayed writes (refused by the contract)	
content changed after signing	0 / 100 accepted
signed with another agent's key	0 / 100 accepted
signed with an unregistered key	0 / 100 accepted
replayed under another record id	0 / 100 accepted
replayed from another deployment	0 / 100 accepted
substituted model	0 / 100 accepted
sent by a non-owner account	0 / 100 accepted
correctly signed (control)	100 / 100 accepted
History rebuilt by the operator	
altered records, amendment-5 check	0 / 200 detected
registrations against anchor	10 / 10 differ
altered records against anchor	200 / 200 detected
unaltered records against anchor	0 / 29,800 flagged
Honest run	
records, registrations, anchors	0 tampered, 0 + 0 mismatches

Local nodes, 30,000 records; the anchor chain is a second local node standing in for an external chain. Batches of 100: 126,005 gas per record (amendment 5: 95,912); anchoring 113 gas per record; inclusion proof 320 bytes.

The hardened variant was tested on the same 30,000 records, with the anchor chain simulated by a second local node (Table 6). The contract refused every forged or replayed write: content changed after signing, signatures by another agent's or an unregistered key, signatures replayed under another record identifier or from another deployment, and substituted models (100 attempts each), while all 100 correctly signed control records were accepted. When the operator rebuilt the entire history with 200 altered records, per-record verification found nothing, as expected when chain and off-chain store are both replaced; the registrations exposed the operator's keys for all ten agents, and the anchors identified exactly the 200 altered records, whichever field was changed, with no unaltered record flagged. The hardening raised the cost to about 126,000 gas per record in batches of 100 (from 96,000), plus 113 gas per record for anchoring. These results are descriptive and rest on the assumption that the anchor chain cannot be rewritten.

5.5 Cross-domain comparison and sensitivity

Table 7 Cross-domain comparison (H6: same verdict in both domains) and robustness of the classical verdicts across β, ρ and ω

Agents	Outcome	Pharmaceutical	Energy	Same verdict
Classical	Fidelity F	+83.5% (improved)	+3.6% (negligible)	no
	Evidence support E	+75.7% (improved)	+4.3% (negligible)	no
	Manipulation M	-69.7% (improved)	-44.4% (improved)	yes
LLM	Fidelity F	-1.9% (negligible)	+3.4% (negligible)	yes
	Evidence support E	-1.0% (negligible)	-0.3% (negligible)	yes
	Manipulation M	-22.0% (improved)	-12.6% (improved)	yes
Sensitivity (pharma)	verdicts changed	none (all values of β, ρ, ω)		
Sensitivity (energy)	verdicts changed	ω: fidelity, evidence		

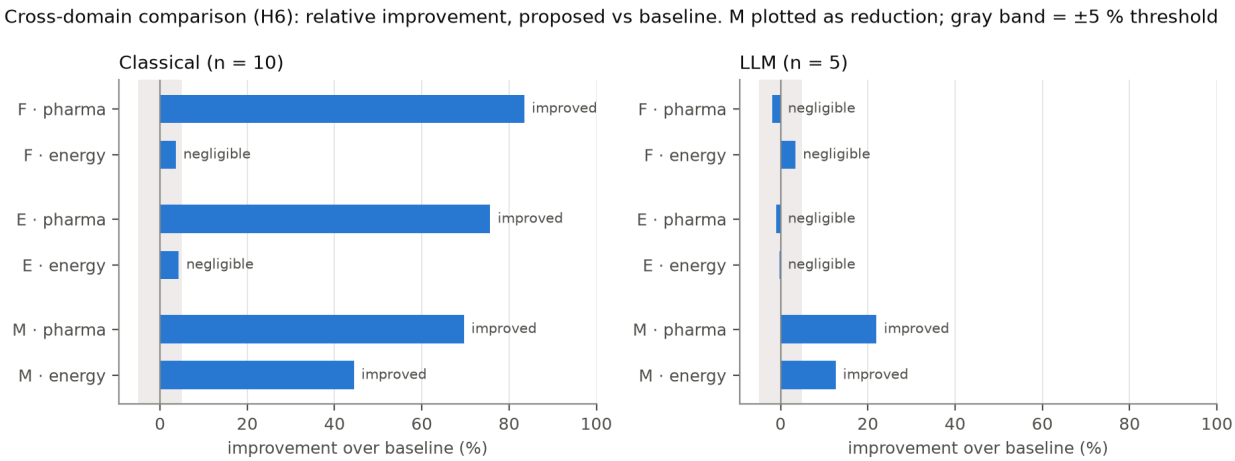


Fig. 8 Relative improvement of the proposed system over the baseline in each domain; manipulation is plotted as a reduction; the grey band marks the 5% threshold

With classical agents, both domains give the same verdict only for manipulation, so H6 holds for one of three outcomes (Table 7, Figure 8). The pharmaceutical verdicts are unchanged at every value of β, ρ and ω. In energy they are unchanged across β and ρ, but fidelity and evidence support change from negligible to improved at $\omega=0.75$ (*+27.8%* and *+28.6%*): when steering the decision is worth more, the baseline's explanations deteriorate and the mechanism's advantage appears. In pharma the fidelity gain persists at $\beta=0$ (*+74.7%*), so it is produced by the evidence and manipulation terms together and should not be attributed to the fidelity weight alone.

5.6 LLM agents

Table 8 LLM agents: 2×2 wording × payoff experiment on mean stated confidence ($n=5$ seeds; Holm over the three effects; added after the main analysis)

Domain	Effect	Δ [95% CI]	p (Holm)	Verdict
pharma	wording	-0.299 [-0.306, -0.291]	*<0.0001*	lower
	payoff	+0.001 [+0.000, +0.002]	0.0971	no detectable effect
	interaction	-0.004 [-0.008, -0.000]	0.0971	not detected
energy	wording	-0.164 [-0.167, -0.161]	*<0.0001*	lower
	payoff	-0.000 [-0.002, +0.002]	0.9884	no detectable effect
	interaction	+0.006 [+0.004, +0.009]	0.0046	detected

With LLM agents (Table 7), manipulation fell in both domains (*-22.0%* and *-12.6%*, Holm $p<0.0001$) while fidelity and evidence support were negligible; fidelity stayed between 0.45 and 0.46 in every system; sample replies show the model misreading the signs and levels in its own tool output. The fall in manipulation came mostly from lower stated confidence: the confidence-inflation component accounts for *-0.060* of the *-0.065* change in pharma and *-0.029* of *-0.035* in energy. To find its cause we added, after the main analysis, a 2×2 experiment that crosses the wording of the reward rules in the prompt with the payoff actually applied (Table 8). The wording lowered confidence by 0.299 in pharma and 0.164 in energy, whereas no payoff effect was detected in either domain; the energy run, specified before it was carried out, reproduced the pharmaceutical pattern, with a small wording-by-payoff interaction (*+0.006*) in energy only. For these agents the behavioural change reflects what they are told, not what they are paid.

6 Discussion

What the mechanism does. Putting audited explanation quality into the utility reliably reduced manipulation: in both domains, with classical and LLM agents, in the ablation without reputation and across the sensitivity range. It also removed the advantage of false explanations: in the pharmaceutical case the agents that still deviate do so with truthful explanations. Whether explanation quality itself improves depends on how much the baseline leaves to gain. In pharma, where unilateral manipulation pays under a decision-only utility, the gains were large; in energy, where honesty was already a best response, they were significant but small, and they became large only when the private stake ω rose.

What it does not do. The mechanism did not make the collective decision more robust to a growing share of agents that always manipulate. It scores reports; it cannot change the votes of agents that ignore the payoff, and reputation reacts only after several rounds. Defences against persistent attackers need another instrument, such as exclusion thresholds or robust aggregation. With LLM agents the improvement cannot be credited to the payoff: a prompt that states the rules produces the same change. Designers of incentive mechanisms for LLM agents should therefore test whether agents respond to payoffs at all before attributing behaviour to them.

Security scope. The evaluated threats are dishonest agents with valid identities (false attribution, inflated confidence, unsupported evidence, strategic voting, collusion of two), changes to recorded artefacts, and forged, replayed or rewritten records. Out of scope are attackers who adapt to the utility, Sybil identities and whitewashing (a new identity starts at reputation 0.5), poisoned source data, models built to mislead SHAP, a declared model that differs from the one actually run, stolen agent keys, and network-level attacks; all components ran in one process on one machine. The provenance layer is a research prototype, not a system hardened against infrastructure attacks.

Limitations. Both reference decisions are constructed, and decision accuracy means agreement with them, not with real choices. The pharmaceutical data cover one programme's antiretroviral and HIV-test deliveries, and the candidate sets are inferred. The energy battery parameters are simulated. Fidelity is measured against SHAP of the agent's own model and therefore needs white-box access; it inherits the limits of SHAP and says nothing about whether the model itself is right. The terciles that define evidence levels and the incentive weights were set a priori; other choices would give other numbers. Reputation creates path dependence. Inference rests on ten seeds (five for LLM agents) on one dataset per domain. The LLM results come from one 7-billion-parameter model, and the factorial experiment was added after the main analysis, as was the hardened provenance layer, whose anchor chain was simulated by a second local node. Overhead was measured on local nodes and will differ on production networks.

7 Conclusion

We proposed an explanation-aware incentive for multi-agent AI decisions, in which audited explanation fidelity, evidence support and manipulation enter each agent's utility, reputation weights votes and a blockchain records the artefacts, and evaluated it under a pre-registered protocol on real pharmaceutical and energy data. The incentive reduced strategic manipulation in both domains and removed the profit of false explanations; it improved explanation quality strongly only where the baseline left room; it did not increase robustness to malicious agents; and the provenance layer detected every tampered record and, with agent signatures and anchoring, refused every forged write and exposed a history rebuilt by its operator. For LLM agents, the effect followed the wording of the rules rather than the payoff. Future work should test the mechanism with black-box fidelity estimates, real decision outcomes, larger and more diverse LLMs, and defences that act on persistent attackers.

Declarations

Funding. [To be completed on the title page.]

Competing interests. [To be completed on the title page.]

Data availability. The pharmaceutical data are the public USAID SCMS delivery history (Development 2015); the energy data are available from Mendeley Data (García-Muñoz 2025). Processed scenarios are regenerated deterministically by the code.

Code availability. The code, configuration files, frozen protocol with its amendments, per-experiment summaries and the scripts that generate every table and figure will be released in a public repository on acceptance; an anonymised copy is available to reviewers.

Author contributions. [To be completed on the title page.]

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